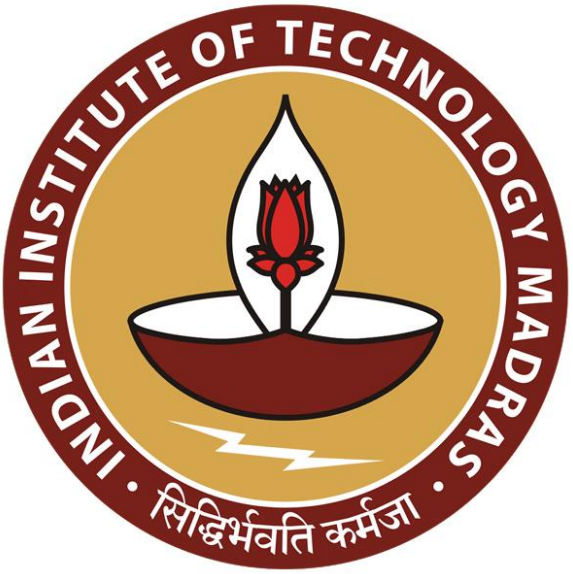
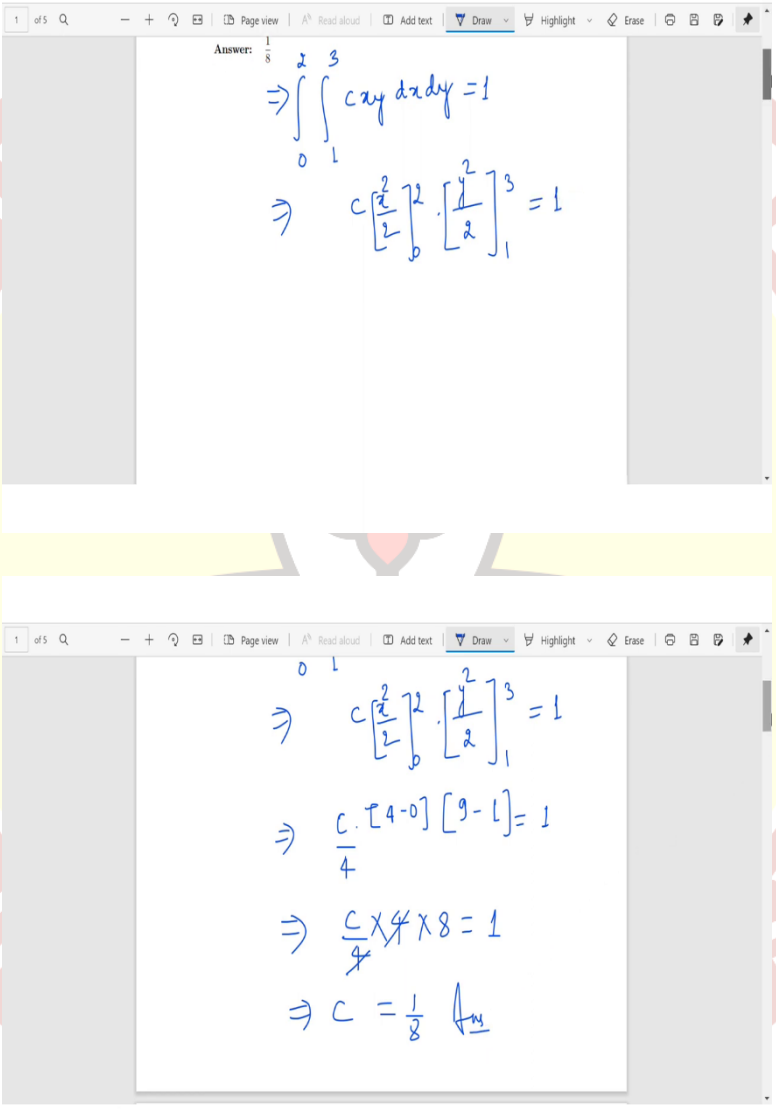


IIT Madras

ONLINE DEGREE

Machine Learning Foundations
Department of Computer Sciences and Engineering
Indian Institute of Technology Madras
Week 11 Turotiral
Professor: Vishal
Examples of Continuous random variables

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Hello, everyone, I am Vishal, and I am your instructor for that MLF course. And today, we will be going through some of the questions, which are related to weak 11. Maybe that will help you for some conceptual clarity, and also will help you in solving some of the questions in the graded assignments and practice assignment. So, let us get started.

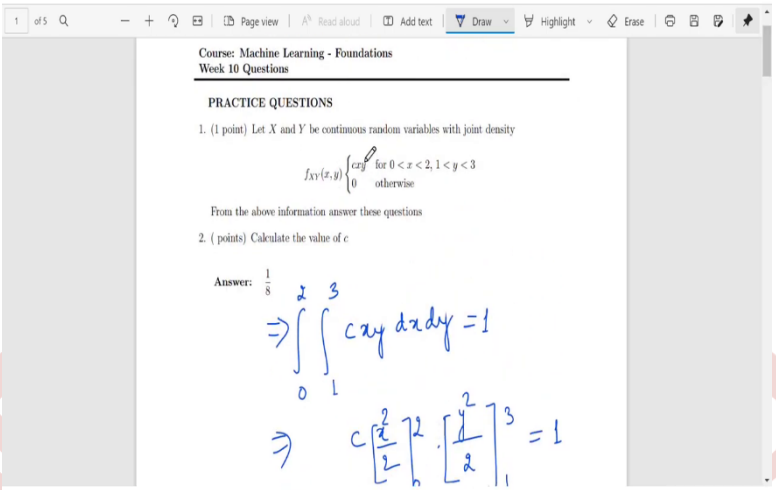
So, we will go through, this our first question. So, in this question, here it has been given. So, we have x and y with a continuous random variable with joint density function as given here, fxy is equal to cxy where x is defined from 0 to 2 and y is defined from 0 to sorry, 1 to 3. And for any other cases, the probability density function is 0. So, from the above information we need to solve, solve the question, some of the problem.

So first, we have to calculate the value of c. So that is pretty simple, you already everyone know, just we need to integrate the joint density function and that will be equal to 1, the integral will be equal to 1. So, from there, we can get the value of c. So, what we will be doing? Here we have, so we have this functional cxy or whether we are integrating dx dy or dy dx, that will not make any difference.

So, the value of x is from 0 to 2 and the value of y is from 1 to 3. And that summation will be equal to 1, right, that is one of our basic you can say the function or the probability density function, then the summation will be equal to 1. So, now what we will do, you just have to do the integral integration. You can directly do the integration c, that is a constant for x square by 2 and that is from 0 to 2 and for y, that is y square by 2 and that is from 1 to 3.

And that summation will be equal to 1. So when we will do the calculations c into what is this, 2 chi square that is 4 minus 0 and directly 2 into 2 we can directly write it here 4. And now, 3 chi square is 9 minus 1 chi square is 1, so is equal to 1. So, c by 4 into 4 into 8 equal to 1, 4, 4 cancel, and then c will be equal to 1 upon 8, that is our value of c. So, now, we will move to the second question.

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Course: Machine Learning - Foundations
Week 10 Questions

PRACTICE QUESTIONS

1. (1 point) Let X and Y be continuous random variables with joint density

$$f_{XY}(x,y)\begin{cases} cxy & \text{for } 0<x<2, 1<y<3 \\ 0 & \text{otherwise} \end{cases}$$

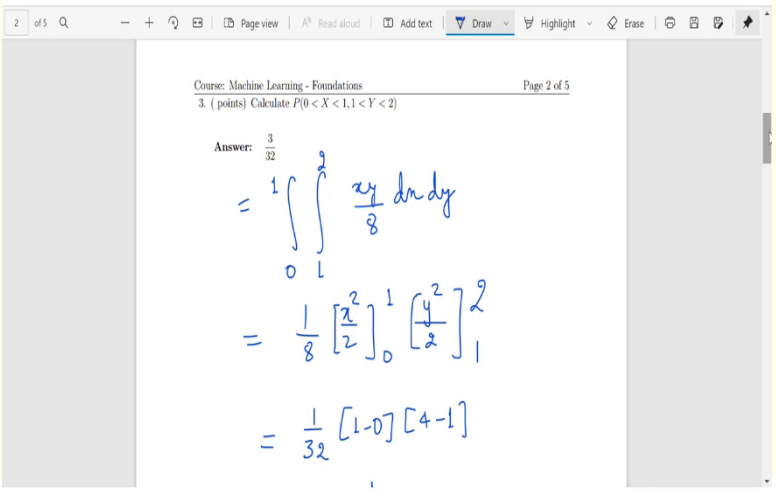
From the above information answer these questions

2. (points) Calculate the value of c

Answer: $\frac{1}{8}$

$$\Rightarrow \int_0^2 \int_1^3 cxy\,dx\,dy = 1$$

$$\Rightarrow c\left[\frac{x^2}{2}\right]_0^2 \cdot \left[\frac{y^2}{2}\right]_1^3 = 1$$



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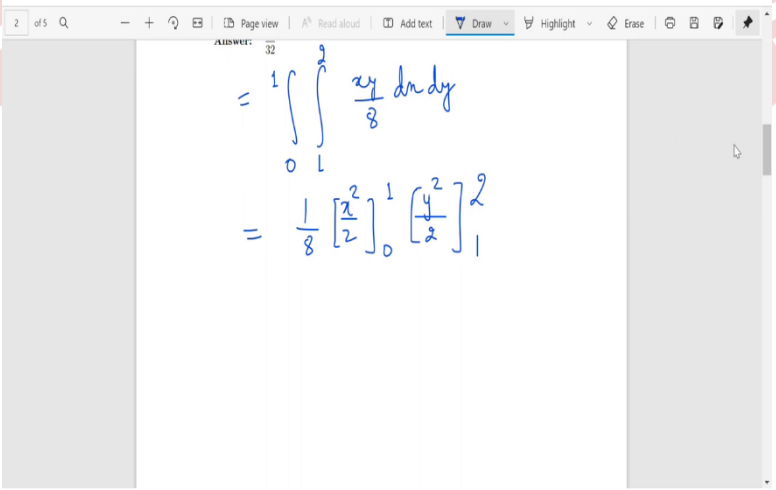
3. (points) Calculate $P(0<X<1, 1<Y<2)$

Answer: $\frac{3}{32}$

$$= \int_0^1 \int_1^2 \frac{xy}{8}\,dx\,dy$$

$$= \frac{1}{8}\left[\frac{x^2}{2}\right]_0^1 \left[\frac{y^2}{2}\right]_1^2$$

$$= \frac{1}{32}[1-0][4-1]$$



Answer: $\frac{3}{32}$

$$= \int_0^1 \int_1^2 \frac{xy}{8}\,dx\,dy$$

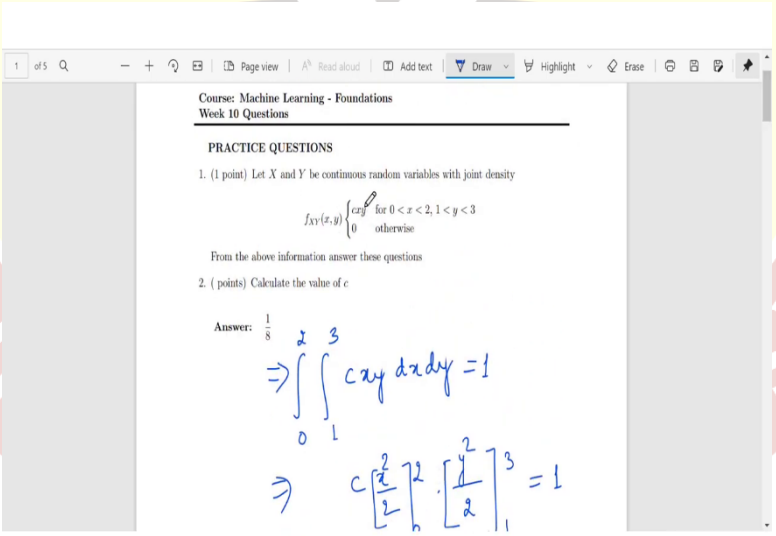
$$= \frac{1}{8}\left[\frac{x^2}{2}\right]_0^1 \left[\frac{y^2}{2}\right]_1^2$$

The data would be same for all these questions. Here we need to calculate the probability for the probability whenever the x will vary from 0 to 1 and y will vary from 1 to 2, in the same joint probabilities, joint density function, so now we are having the value of c. The joint density function would be xy upon 8, because c already we have got as 1 upon 8. So, here, similar type of questions we just have to integrate.

Here we need to integrate x from 0 to 1, because they are asking the probability whenever the x is lying between 0 to 1 and y is lying between 1 to 2. 0 to 1 and the y will be from 1 to 2. And the fxy is here, xy upon 8 and dx dy. So, here, we can do the same thing just we need to integrate it, x square by 2, x will vary from 0 to 1 and y square by 2 and y will lye from 1 to 2.

So, there 2 into 2 is 4, 1 upon 32, x1 minus 0 multiplied with y2 chi square is equal to 4 minus 1. So, the value would be 4 minus 3 is 3, so, 3 by 32 would be the value. So, for this the probability whenever the x will vary from 0 to 1 and y will vary from 1 to 2 would be 3 upon 32, so, we are getting this.

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4. (points) Calculate $P((X+Y)<3)$

$$= \int_{x=0}^{2} \int_{y=1}^{3-x} f_{xy}(x,y)\,dx\,dy \quad \text{or}$$

$$\int_{y=1}^{3} \int_{x=0}^{1} f_{xy}(x,y)\,dx\,dy$$

$$\Rightarrow \frac{1}{2\times 8} \int_0^2 x\,dx\,[(3-x)^2 - 1]$$

$$\Rightarrow \frac{1}{16} \int_0^2 \left[(9-6x+x^2-1)x\,dx\right]$$

$$\Rightarrow \frac{1}{16}\left[\frac{x^4}{4} - 6\frac{x^3}{3} + 8\frac{x^2}{2}\right]_0^2$$

$$\Rightarrow \frac{1}{16}[4 - 16 + 16]$$

$$\Rightarrow \frac{1}{4} = 0.25$$

4. (points) Calculate $P((X+Y)<3)$

$$= \int_{x=0}^{2} \int_{y=1}^{3-x} f_{xy}(x,y)\,dx\,dy \quad \text{or}$$

$$\int_{y=1}^{3} \int_{x=0}^{1} f_{xy}(x,y)\,dx\,dy$$

$$\Rightarrow \frac{1}{} \int_0^2 x\,dx \left[\frac{y^2}{}\right]_{}^{3-x}$$

Now, we will move to the third question. So, here we have been given we need to calculate the probability of x plus 5 is less than 3, so here, x and y are dependent on each other. So, this can be solved with two, three methods. So, I will be one method is pretty simple, you just have to do the integration from you can do the integration like you can vary the x is from 0 to 2 and then y you can vary from 1 as the sum for both these both this random variable 3, so if we are putting y as 1 because the y will start from 1 as given in the data y is starting from 1.

So, y will be starting from 1, so here, it cannot be more than 3 because x plus y is less than 3. So, we can write it there 3 minus x dx dy. So, either you can do with this or you can do from this formula. You can just put the value of y, y will vary from 1 to 3 and x will vary from 0 to 2 and f and here we can write directly the joint density function dx dy.

So, we can calculate from both this formula, from both these equations we will be getting the same value for the probability x plus y less than 3. So, I will be calculating from here, you can as a homework you can calculate from this way also and you will be getting the same value. There are other methods also with a graph and all we can solve this, but I am like solving with these equations only.

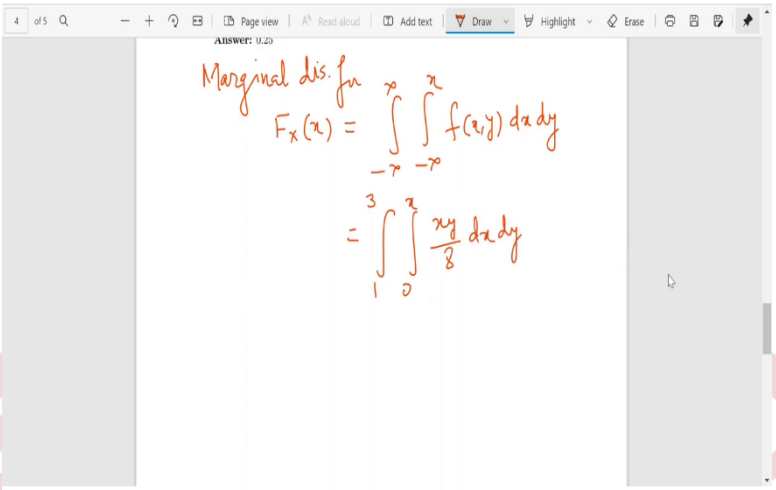
So, with this equation, I will be doing the calculation here. Fxy is for xy upon 8 so 1 upon 8 I will write directly. And then for the integration we already know here it is xy upon 8, so 1 by 8 because this is a constant we can write it outside. So, now we have integration of xdx and I have done the integration of y, so y what would be the integration for y, y square upon 2. And what would the valuable vary from 1 to 3 minus x and x will vary from 0 to 2.

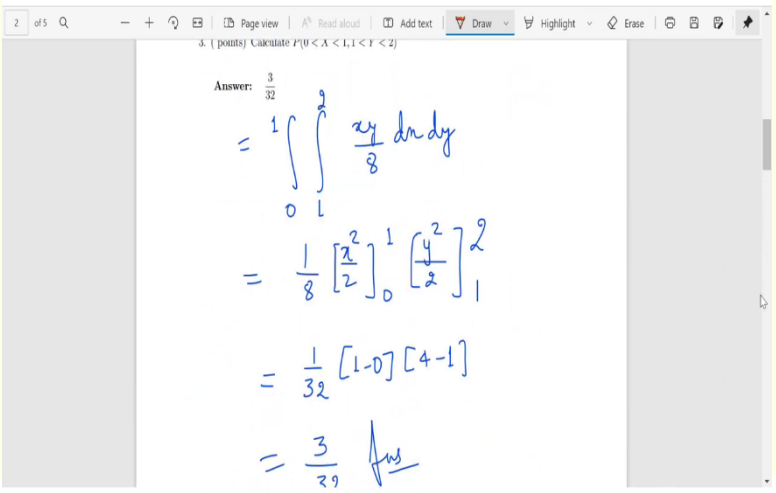
So now, 1 upon 8 is fine and now this is the integration xdx. Here we can write it directly 3 minus x whole square and 2 we can write it here into 2. So, this 2 we can write it here. 3 minus x whole square minus 1 whole square. Now, we have 1 upon 16 we can just open this a minus b whole square we can just open it, and then after we can integrate it, so first we will open it. So, this will be 9 minus 6x plus x square minus 1 and we have to multiply this with x and then we have to integrate this. We have to integrate from 0 to 2.

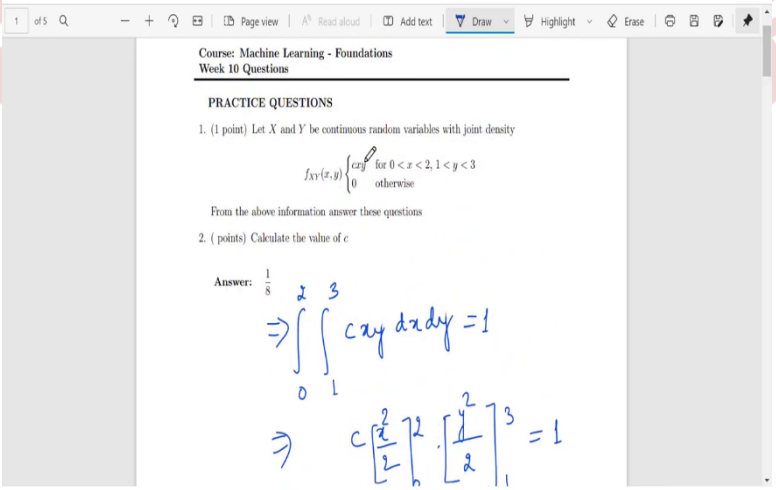
So, what we will do, we will just integrate this 1 upon 16. So, if we will integrate this x square into x, that is x cube, so the integral would be x4 upon 4. Similarly, for this one 6x into x square, so x square, sorry, x into x, x square so x cube by 3, plus 9 minus 1, 8 so 8x square by 2, and we just have to put the value as we have to integrate from 0 to 2, so we will just write it like this.

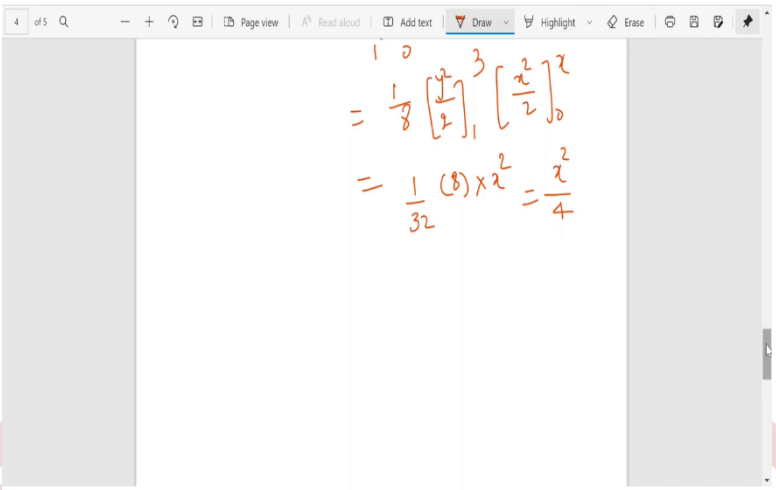
So, 1 upon 16 so what is this, we will put the value 2, so 2 to 2 are 4, 16 upon 4, that is 4 minus 2 cube into 2, 16 plus 4 into 4, 16 so this 16 will get cancelled. So, we will be having 1 upon 4, that is, 0.25. So, similar thing we can do with this also this. So, we can do with this also. So, we will be getting the same value, as we have calculated from this equation. So, now we will move to the next question.

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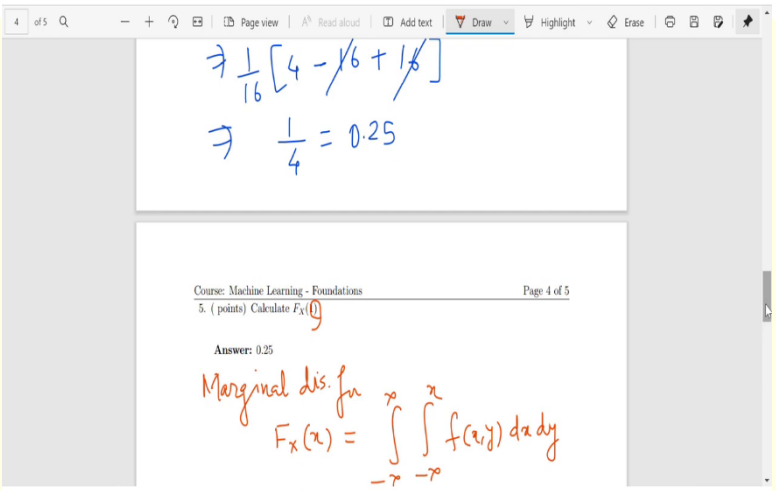






$$= \frac{1}{8}\left[\frac{y^2}{2}\right]_1^3 \left[\frac{x^2}{2}\right]_0^2$$

$$= \frac{1}{32}(8)x^2 = \frac{x^2}{4}$$



$$\Rightarrow \frac{1}{16}[4 - 6 + 16]$$

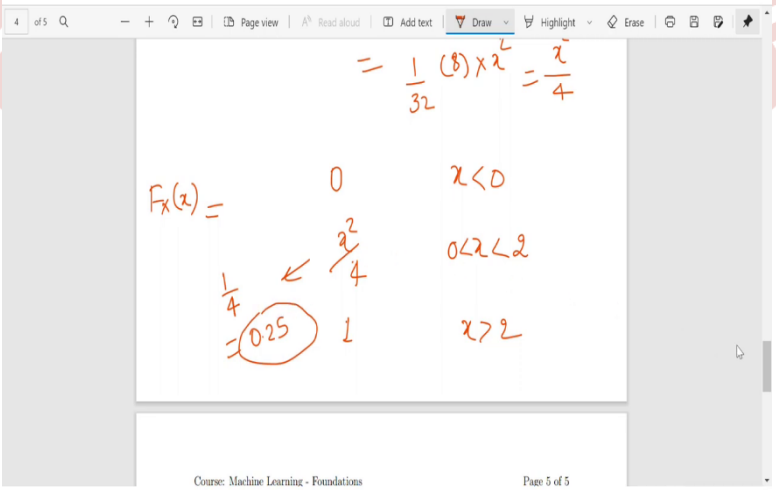
$$\Rightarrow \frac{1}{4} = 0.25$$

5. (points) Calculate $F_X(1)$

Answer: 0.25

Marginal dis. fn

$$F_X(x) = \int_{-\infty}^{\infty}\int_{-\infty}^{x} f(x,y)\,dx\,dy$$



$$= \frac{1}{32}(8)x^2 = \frac{x^2}{4}$$

$$F_X(x) = \begin{cases} 0 & x<0 \\ \frac{x^2}{4} & 0<x<2 \\ 1 & x>2 \end{cases}$$

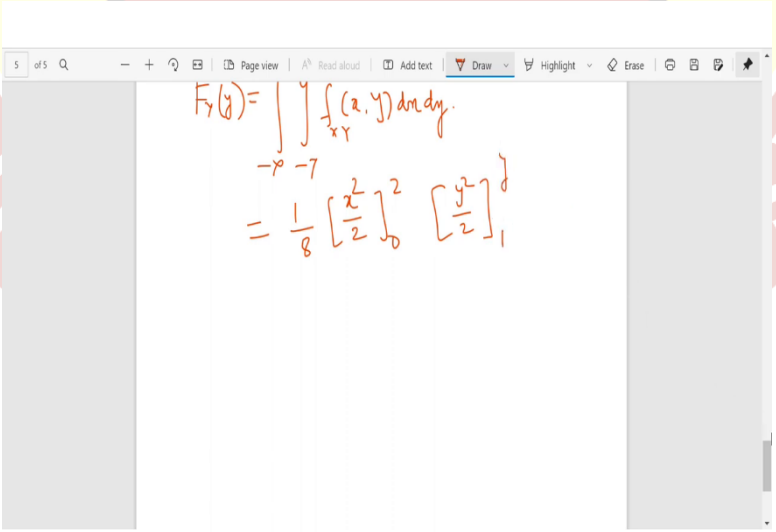
$\frac{1}{4} = 0.25$

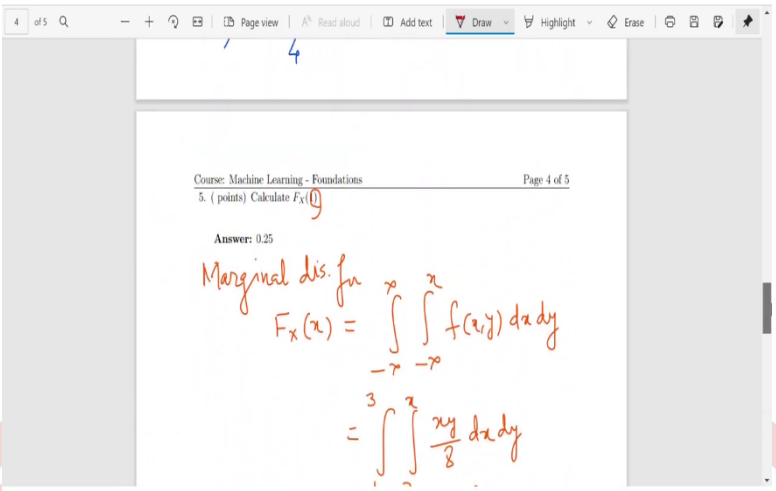
Next question, we just have to calculate the marginal distribution for x. So, for marginal distribution of x, we have to calculate, we already know the formula for marginal distribution of x for marginal distribution we can define this Fx, x and here, the y will vary from minus infinity to infinity and the x will vary from minus infinity to x and this will be the joint distribution function.

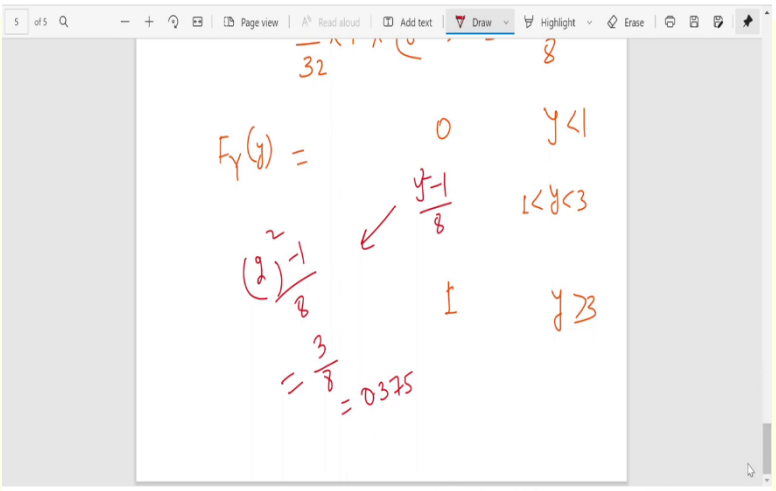
But here, y is varying from 1 to 3. Twice then from 1 to 3, and x is varying from 0 to 2. So, x will be from 0 to x. And from joint distribution function we already have xy upon 8, and dx into dy. So, we just have to do the integration and that is it, we will get the value. So, 1 upon 8 y square upon 2, the value from 1 to 3, and x would be from x square by 2, x will be from 0 to x and we will get the function as 1 upon 32, and then we will put 3 here so we will get 3 chi square 9 minus 1 that is 8 into x square.

So, we will be having x square upon 4. So marginal distribution function how generally we define? Fx, so for x less than 0, this density function will be 0, because the density function, this marginal density would be defined from x equal to 0 to 2 for the x less than 0, it would be 0. And for x greater than 2, it would be 1 and for the x between 0 to 2, we have calculated x square by upon 4. So, we have to calculate for the value of 1. So, for 1, we will just put the value as 1 here. So, after we put 1, 1 upon 4, so that is 0.25 will be the answer.

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So, now we will move to the next one, and the next one is also similar to whatever we have solved earlier. This is a marginal distribution function of for the y. So, how will calculate for the marginal distribution of y, x will be from minus infinity to infinity and y will be from minus infinity to y, and the function of the Fx, dx dy. So, we have already the function 1 upon 8 so 1 upon 8 can be outside and here x square upon 2, x will be from 0 to 2, y square upon 2, y will be from 1 to y, and just have to do the integration.

So, 1 upon 8, 8 into 2 into 2, so 1 upon 32 and the value for we will put the value as 4 and here y would be y square minus 1. So that is y square minus 1 upon 8. So here in the similar way, as we did for the last one, so the marginal distribution for y would be, so for y less than 1 it will be 0 and for y greater than 3, it would be 1, that we did. And then for the y between 1

and 3, it would be y, sorry, y square one second, so for y between 0 to 1, it would be y square minus 1 upon 8. So, for which value we need to calculate.

So, we have to calculate for y equal to 2. So, we will just put the value of y equal to 2, so 2 chi square minus 1 upon 8 that is 4 minus 1, 3 upon 8 that is 0.375. So, we have done with all the questions. Hopefully, you would have gained something from this and will help you in solving the questions related to credit and practice assignments. So, thank you. Thanks a lot.

